Design of Unbalanced Mach–Zehnder Interferometers for Waveguide Group Index Extraction

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***Abstract*—This report designs, models, and experimentally characterizes unbalanced silicon Mach–Zehnder interferometers (MZIs) to extract waveguide group index from measured free spectral range (FSR). We define waveguide geometry, simulate guided TE modes, and compute $n_{\text{eff}}(\lambda)$ and $n_g(\lambda)$. A compact polynomial model and an analytical MZI transfer function are used to predict transmission spectra across design variants spanning waveguide width and path imbalance ΔL. Fabricated devices are measured, and FSR is extracted from transmission spectra showing consistent inverse scaling with ΔL and good agreement with simulation. n_g is then robustly extracted using a linear fit approach. At 1550 nm, the extracted group indices are approximately 4.4, 4.2, and 4.1 for widths of 400, 500, and 600 nm, respectively. A comparison of measurement at Scylla and Iris setups is also provided.**

I. Introduction

Silicon photonics enables dense, CMOS-compatible integration of optical waveguides and passive components on silicon-on-insulator (SOI), supporting scalable transceivers and wavelength-division multiplexing systems with compact footprints and high-volume manufacturability [1]. Accurate knowledge of waveguide dispersion—especially the group index $n_g(\lambda)$—is essential for predicting interferometer periodicity, phase sensitivity, delay, and bandwidth in circuits such as filters, sensors, and coherent links [2]. However, $n_g(\lambda)$ depends on waveguide geometry and fabrication variations, motivating an experimental extraction method that can be directly compared to simulation.

The objective of this design proposal is to design an unbalanced interferometer whose transmission fringes provide a direct measurement of $n_g(\lambda)$ via the free spectral range (FSR). We will (i) select waveguide widths, (ii) simulate TE mode profiles and compute $n_{\text{eff}}(\lambda)$ and $n_g(\lambda)$, (iii) develop compact dispersion models for circuit-level prediction, and (iv) choose practical values of ΔL and waveguide width such that the resulting FSR is resolvable with the available measurement setup. The final outcome is a set of MZI design variants and a validated simulation-to-extraction workflow to be applied to measured spectra.

II. Theory

A Mach–Zehnder interferometer (MZI) splits an input optical field into two arms that accumulate different phases and then recombine. For a guided mode with effective index $n_{\text{eff}}(\lambda)$, the propagation constant is

$$\beta(\lambda) = \frac{2\pi}{\lambda}\, n_{\text{eff}}(\lambda). \tag{1}$$

Let the arm lengths be L_1 and L_2, with imbalance

$$\Delta L = L_2 - L_1. \tag{2}$$

The differential phase is

$$\Delta\phi(\lambda) = \beta_2(\lambda)L_2 - \beta_1(\lambda)L_1 + \phi_{\text{off}}, \tag{3}$$

where ϕ_{off} captures any constant phase offset (e.g., splitter phase or small asymmetries). In this proposal both arms use the same cross-section and materials and differ only in length, so

$$\beta_2(\lambda) = \beta_1(\lambda) = \beta(\lambda), \tag{4}$$

and therefore

$$\Delta\phi(\lambda) = \beta(\lambda)\Delta L + \phi_{\text{off}}. \tag{5}$$

For an ideal MZI with 50/50 splitters and negligible loss imbalance, the normalized transmission is

$$T(\lambda) = \frac{1}{2}\left[1 + \cos\big(\Delta\phi(\lambda)\big)\right]. \tag{6}$$

More generally, unequal arm amplitudes reduce contrast but do not change the fringe periodicity:

$$T(\lambda) = T_0 + T_1 \cos\big(\Delta\phi(\lambda)\big). \tag{7}$$

In the following derivation we neglect loss and set ϕ_{off} to zero (or absorb it into the integer fringe index).

Transmission maxima occur when

$$\Delta\phi(\lambda_m) = 2\pi m, \qquad m \in \mathbb{Z}, \tag{8}$$

so that

$$\beta(\lambda_m)\Delta L = 2\pi m. \tag{9}$$

For adjacent maxima at λ_m and $\lambda_{m+1} = \lambda_m + \Delta\lambda$ (with $\text{FSR}_\lambda \equiv \Delta\lambda$),

$$\big[\beta(\lambda_{m+1}) - \beta(\lambda_m)\big]\Delta L = 2\pi. \tag{10}$$

For small $\Delta\lambda$,

$$\beta(\lambda_{m+1}) - \beta(\lambda_m) \approx \frac{d\beta}{d\lambda}\Delta\lambda, \tag{11}$$

hence

$$\frac{d\beta}{d\lambda}\,\text{FSR}_\lambda\,\Delta L \approx 2\pi. \tag{12}$$

With $\beta(\lambda) = \frac{2\pi}{\lambda} n_{\text{eff}}(\lambda)$,

$$\frac{d\beta}{d\lambda} = -\frac{2\pi}{\lambda^2}\left(n_{\text{eff}} - \lambda \frac{dn_{\text{eff}}}{d\lambda}\right) = -\frac{2\pi}{\lambda^2} n_g(\lambda), \quad (13)$$

where

$$n_g(\lambda) = n_{\text{eff}}(\lambda) - \lambda \frac{dn_{\text{eff}}}{d\lambda}. \quad (14)$$

Using magnitudes,

$$\text{FSR}_\lambda \approx \frac{\lambda^2}{n_g(\lambda)\Delta L}, \quad (15)$$

and therefore

$$n_g(\lambda) \approx \frac{\lambda^2}{\text{FSR}_\lambda \, \Delta L}. \quad (16)$$

This first-order approximation assumes (i) dispersion does not vary significantly across one fringe period, (ii) ΔL is accurately known from layout, and (iii) the observed fringes correspond to a single guided mode.

III. Modelling and Simulation

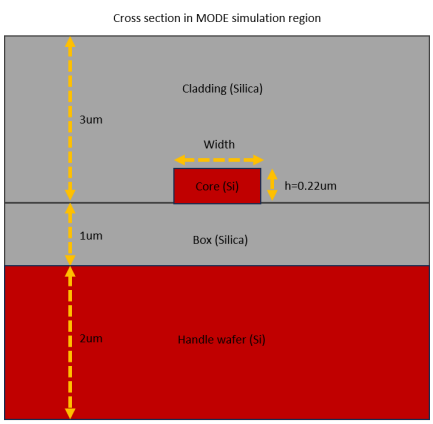

Figure 1. Waveguide cross-section geometry: silicon core thickness $h = 220$ nm and width $w \in \{400, 500, 600\}$ nm.

This section summarizes the modeling workflow and simulation results used to predict the dispersion and spectral response of the designed interferometer circuits prior to measurement. We begin by defining the waveguide and circuit geometries and numerically extracting the wavelength-dependent effective index and group index, $n_{\text{eff}}(\lambda)$ and $n_g(\lambda)$, which are plotted with respect to wavelength. The simulated dispersion is then reduced to a compact polynomial waveguide model that enables efficient circuit-level calculations. Using this compact model, we formulate the device transfer function(s) and simulate the expected transmission spectrum(s) for one or more photonic circuits. A DOE (design of experiment) table is included to document the design splits (e.g., waveguide widths and path-length imbalance ΔL) and to quantify how the predicted free spectral range (FSR) changes with ΔL and waveguide widths.

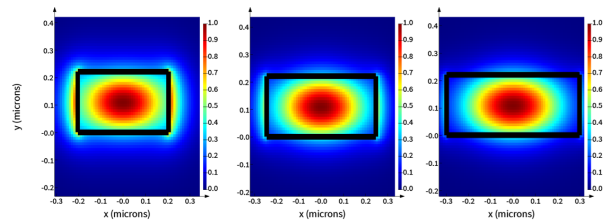

Figure 2. Simulated fundamental waveguide mode profile from Lumerical MODE for width $w \in \{400, 500, 600\}$ nm.

Figure 1 summarizes the assumed SOI strip-waveguide cross section used throughout the simulations, with fixed silicon thickness $h = 220$ nm and three candidate widths $w \in \{400, 500, 600\}$ nm. This geometric definition is the starting point for the eigenmode calculations because it sets the modal confinement and the wavelength dependence of the propagation constant. To verify the guided mode used for dispersion extraction, Fig. 2 shows the simulated fundamental mode profile (e.g., $|E(x, y)|^2$) from Lumerical MODE Solutions; this plot is used to confirm the intended polarization, the expected field confinement in the silicon core, and that the simulated mode is physically reasonable before proceeding to wavelength sweeps.

Table I
DOE of waveguide width and MZI path-length imbalance with simulated group index and expected FSR.

w (nm)	ΔL (μm)	n_g	n_{eff}	$FSR(nm)$
400	100	4.397	2.226	5.464
400	200	4.397	2.226	2.732
400	400	4.397	2.226	1.366
500	100	4.205	2.444	5.713
500	200	4.205	2.444	2.857
500	400	4.205	2.444	1.428
600	100	4.066	2.565	5.909
600	200	4.066	2.565	2.954
600	400	4.066	2.565	1.477

The design-of-experiments (DOE) for the interferometer circuits is summarized in Table I. For each width w, three imbalance lengths ΔL are selected to span a practical range of FSR values while remaining compatible with layout and test constraints. The table records the corresponding simulated indices (n_g and n_{eff}) used for prediction and provides a clear bookkeeping reference for which geometries map to which expected spectral periods. In particular, the FSR estimates in the final column (computed via $\text{FSR}_\lambda \approx \lambda_0^2/(n_g \, \Delta L)$ at $\lambda_0 = 1550$ nm) enable a quick sanity check that the chosen ΔL values produce resolvable fringes over the available measurement bandwidth.

To enable fast circuit-level prediction without re-running full-wave mode simulations, we constructed a compact dispersion model for the waveguide by fitting the simulated effective index to a second-order polynomial over the operating wavelength band. Specifically, the numerically extracted $n_{\text{eff}}(\lambda)$ data (from Lumerical MODE Solutions and/or MATLAB) were least-squares fitted using a Taylor expansion about a center wavelength λ_0,

$$n_{\text{eff}}(\lambda) = n_1 + n_2(\lambda - \lambda_0) + n_3(\lambda - \lambda_0)^2 . \quad (17)$$

With this parameterization, the conventional waveguide quantities at λ_0 follow directly as $n_{\text{eff}}(\lambda_0) = n_1$ and $n_g(\lambda_0) = n_1 - n_2\lambda_0$, while the dispersion parameter is given by $D = -2\lambda_0 n_3/c$ (in $\mathrm{s/m^2}$). The resulting compact, differentiable description of waveguide dispersion is then used in the interferometer phase model $\Delta\phi(\lambda) = \beta(\lambda)\Delta L$, enabling efficient transfer-function simulations that closely match the underlying numerical results across the wavelength range of interest.

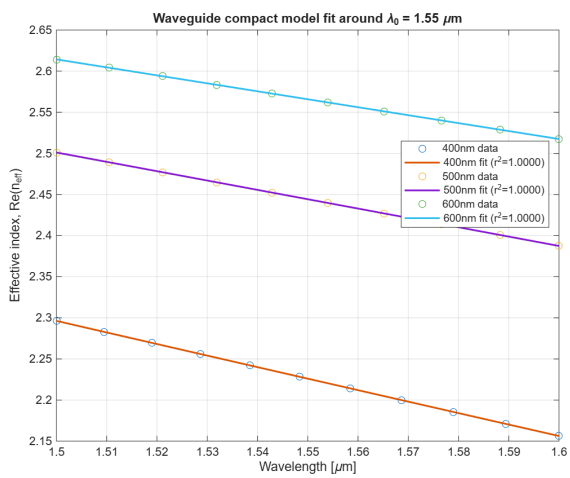

Figure 3. Simulated $n_{\text{eff}}(\lambda)$ (markers) and quadratic compact-model fits (lines) for waveguide widths of 400, 500, and 600 nm, using $n_{\text{eff}}(\lambda) = n_1 + n_2(\lambda - \lambda_0) + n_3(\lambda - \lambda_0)^2$ with $\lambda_0 = 1.55$ μm.

The fitted polynomial coefficients provide a compact summary of the simulated dispersion and are reported in Table. II for each width. In addition to the raw fit parameters (n_1, n_2, n_3), the table lists the derived group index $n_g(\lambda_0)$ and dispersion parameter D at λ_0 to facilitate direct comparison across geometries. Figure 4 visualizes the underlying wavelength sweeps of $n_{\text{eff}}(\lambda)$ and $n_g(\lambda)$, while Fig. 4 (compact-model fit) confirms that the quadratic model captures the simulated $n_{\text{eff}}(\lambda)$ trends with excellent agreement over the wavelength span of interest. These plots serve as both a validation of the compact model and as inputs to the subsequent phase and transfer-function calculations.

Using the extracted dispersion and compact-model coefficients, the wavelength-dependent propagation constant is evaluated as $\beta(\lambda) = 2\pi n_{\text{eff}}(\lambda)/\lambda$, and the interferometer phase difference is computed via $\Delta\phi(\lambda) = \beta(\lambda)\Delta L$. The resulting transfer function(s) are then used to generate predicted transmission spectra for each circuit variant. Figure 5 overlays the simulated transmission responses $T(\lambda)$, enabling a direct visual comparison of how the spectral periodicity (FSR) changes with ΔL and how the waveguide width (through n_g and dispersion) shifts the fringe spacing and spectral alignment. This spectral overlay is particularly useful for selecting a subset of devices that provide well-separated FSRs for

Table II
WAVEGUIDE COMPACT MODEL COEFFICIENTS FROM QUADRATIC FIT $n_{\text{eff}}(\lambda) = n_1 + n_2(\lambda - \lambda_0) + n_3(\lambda - \lambda_0)^2$ (WITH λ IN μM), WITH $\lambda_0 = 1.55$ μM.

	400	**500**	**600**
n_1	2.2260	2.4443	2.5658
n_2 **(μm^{-1})**	-1.4000	-1.1364	-0.9683
n_3 **(μm^{-2})**	1.015×10^{-1}	-1.466×10^{-3}	2.941×10^{-2}
$n_g(\lambda_0)$	4.3960	4.2057	4.0667
D **(s/m^2)**	-1.050×10^{-15}	1.516×10^{-17}	-3.041×10^{-16}

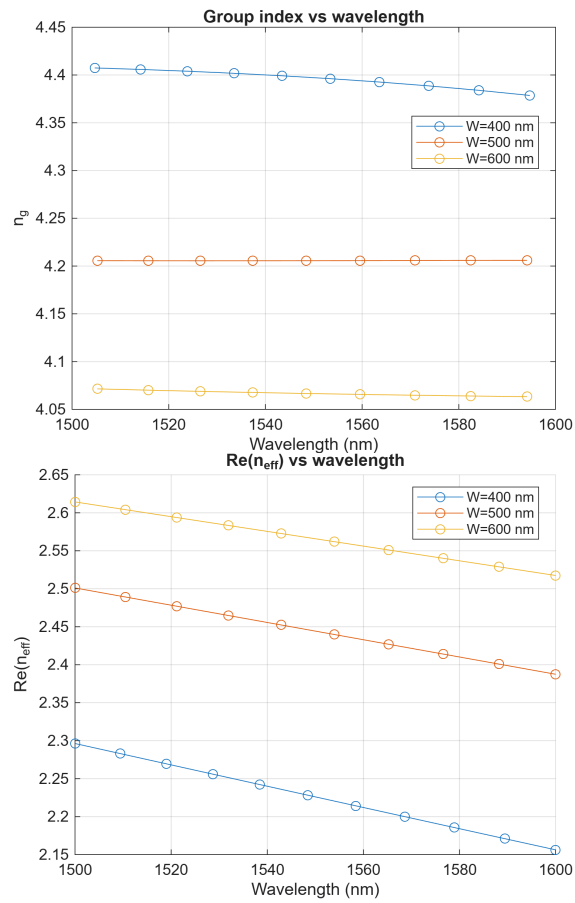

Figure 4. Simulated waveguide dispersion: (top) effective index $n_{\text{eff}}(\lambda)$ and (bottom) group index $n_g(\lambda)$ versus wavelength for width $w \in \{400, 500, 600\}$ nm.

robust group-index extraction in subsequent measurements.

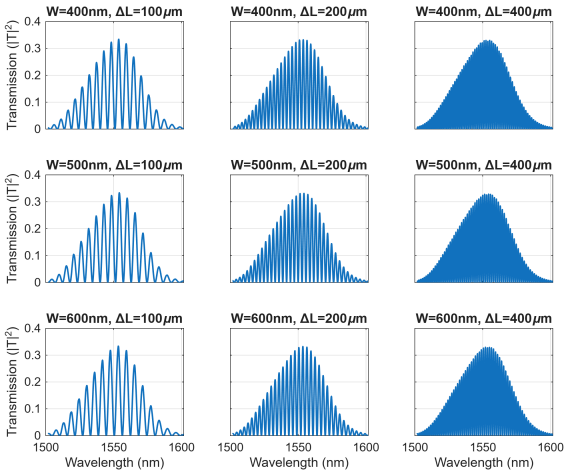

Figure 5. Simulated transmission spectra in Interconnect $T(\lambda)$ for 3 length and 3 width variants. The envelope is a result of the input/output grating couplers

IV. Layout and Fabrication

The final chip layout is shown in Fig. 6, consisting of nine Mach–Zehnder interferometers (MZIs), a calibration waveguide (CALWG), and optional ring resonator structures (RING1 and RING2). The layout is designed to systematically extract the group index by sweeping the arm length imbalance across the MZI structures.

Each MZI incorporates identical grating couplers for optical input and output, ensuring consistent coupling conditions across all devices. The waveguide routing is carefully designed to maintain similar bend radii and minimize excess loss variations between structures. The devices are arranged in a compact array to maximize die area utilization while preserving measurement accessibility.

As summarized in Table III, the MZIs span three waveguide widths (400 nm, 500 nm, and 600 nm) and three arm length differences ($\Delta L = 100$ μm, 200 μm, and 400 μm). This parameter sweep enables systematic evaluation of dispersion effects with respect to both geometric confinement and optical path length. By comparing the spectral responses across these variations, the group index can be accurately extracted over wavelength.

The calibration waveguide (CALWG) is included to characterize the spectral response of the grating couplers and to remove their influence from the measured transmission spectra. This ensures that the extracted interference fringes from the MZIs reflect only the intrinsic waveguide properties.

Optional ring resonator structures (RING1 and RING2) are also included in the layout for potential resonance-based characterization; however, these structures are not analyzed in this work.

Table III
MZI Labels. All TE-polarized.

	waveguide width, arm imbalance
MZI1	$w = 400$ (nm), $\Delta L = 100$ (μm)
MZI2	$w = 400$ (nm), $\Delta L = 200$ (μm)
MZI3	$w = 400$ (nm), $\Delta L = 400$ (μm)
MZI4	$w = 500$ (nm), $\Delta L = 100$ (μm)
MZI5	$w = 500$ (nm), $\Delta L = 200$ (μm)
MZI6	$w = 500$ (nm), $\Delta L = 400$ (μm)
MZI7	$w = 600$ (nm), $\Delta L = 100$ (μm)
MZI8	$w = 600$ (nm), $\Delta L = 200$ (μm)
MZI9	$w = 600$ (nm), $\Delta L = 400$ (μm)

The devices were fabricated using silicon-on-insulator (SOI) wafers with a 220 nm device layer on a 2–3 μm buried oxide, through multi-project wafer (MPW) processes at Applied Nanotools Inc. and the Washington Nanofabrication Facility. Patterning was performed using 100 keV electron-beam lithography [5], with hydrogen silsesquioxane (HSQ) resist, followed by development in tetramethylammonium-based solutions. The waveguide structures were transferred into the silicon layer using inductively coupled plasma reactive ion etching (ICP-RIE) with chlorine-based chemistry. After resist removal and inspection, a $\sim$2–2.2 μm oxide cladding was deposited using plasma-enhanced chemical vapor deposition (PECVD). Standard cleaning, optical and SEM inspection, and thickness verification were performed throughout the process to ensure fabrication fidelity and uniformity.

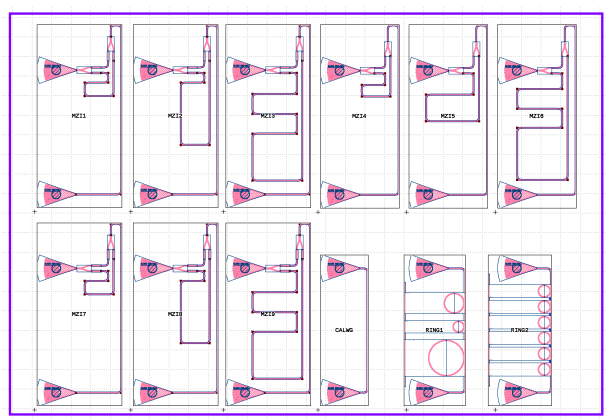

Figure 6. Final layout. MZI Labels explained in Table III. CALWG is the to calibrate grating coupler spectral response. RING1 and RING2 are optional structures which are not discussed in this report.

V. Experimental Data

A. Test setup

This section presents measurement results obtained using two custom-built automated test platforms, Scylla

and Iris. These setups [6, 10] are controlled by Python-based automation software [7], enabling repeatable and high-resolution spectral measurements. An Agilent 81600B tunable laser was used as the input source, while Agilent 81635A optical power sensors were employed for detection. The wavelength was swept with a resolution of 10 pm.

A polarization-maintaining (PM) fiber was used to preserve the input polarization state and efficiently couple TE-polarized light into the grating couplers [8]. For TM grating couplers, a 90° rotation of the input polarization was applied [8]. Light was coupled into and out of the chip using a polarization-maintaining fiber array [9], ensuring stable alignment across all measurements.

B. Calibration

The calibration waveguide (CALWG) was first characterized to isolate the spectral response of the grating couplers and measurement system. The measured transmission spectra from both Scylla and Iris are shown in Fig. 7.

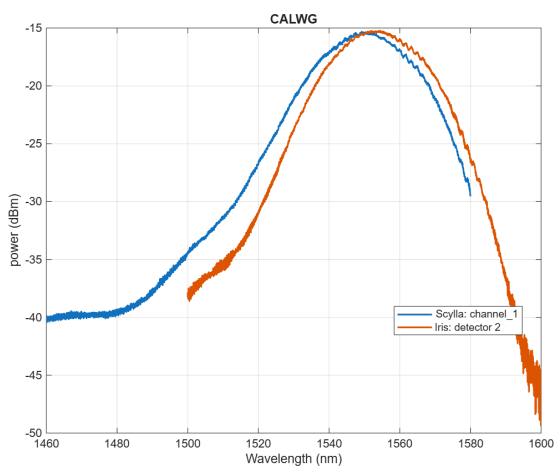

Figure 7. Measured calibration spectra from Scylla and Iris. Both exhibit spectral envelopes dominated by grating coupler response.

Both systems exhibit a similar broadband spectral envelope, confirming consistent grating coupler response and overall system calibration. Small discrepancies in insertion loss and spectral ripple are observed, which can be attributed to differences in fiber alignment, polarization extinction ratio, and system-specific noise characteristics. These variations are slowly varying with wavelength and primarily affect the envelope of the transmission spectrum.

Importantly, the calibration spectra do not exhibit periodic interference features, confirming that any high-frequency spectral fringes observed in subsequent MZI measurements originate from device interference rather than the measurement setup. As a result, the CALWG response can be used as a reference to normalize the MZI spectra, effectively removing the grating coupler envelope and improving the accuracy of free spectral range (FSR) extraction.

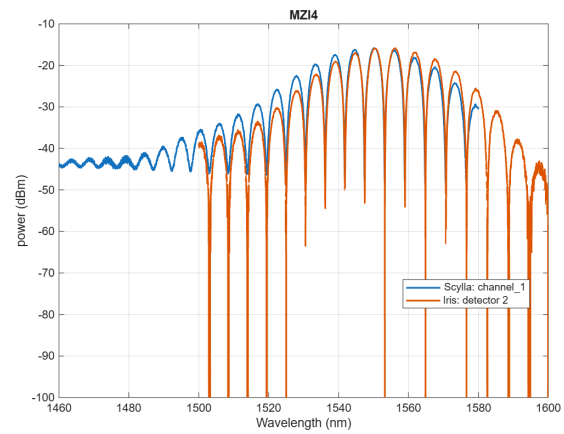

Figure 8. Raw transmission spectra of MZI4 measured on Scylla and Iris. Interference fringes are modulated by the grating coupler envelope.

As an example, the raw transmission spectra of the MZI4 device, with a waveguide width of 500 nm and $\Delta L = 100\ \mu$m, are shown in Fig. 8.

Clear and periodic interference fringes are observed in both measurement systems, confirming proper operation of the Mach–Zehnder interferometer. The fringe spacing is consistent between Scylla and Iris, indicating good agreement in wavelength calibration and measurement repeatability.

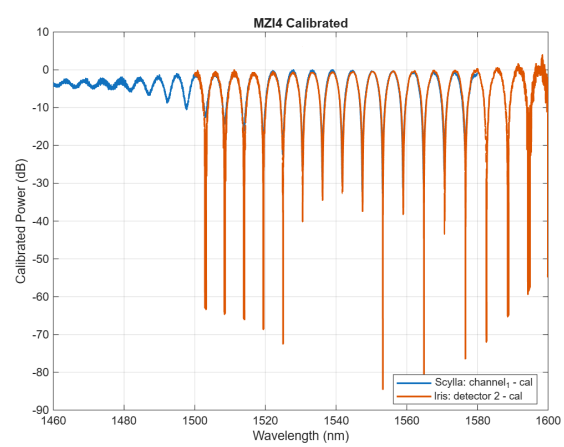

Figure 9. Calibrated spectra of MZI4 after removing the grating coupler response. The interference fringes become more uniform, enabling accurate FSR extraction.

The measured transmission is modulated by a slowly varying grating coupler envelope, which obscures the intrinsic interference fringes and must be removed for accurate free spectral range (FSR) extraction. The raw MZI spectra are therefore normalized using the calibration waveguide response, effectively removing the

envelope and system-dependent variations. This normalization yields a flattened spectrum, shown in Fig. 9, with more uniform fringe visibility, enabling reliable peak identification and accurate extraction of FSR and the corresponding waveguide group index.

C. Extracting FSR vs. wavelength from test data

The calibrated spectra with detected dip locations from Scylla and Iris are shown in Fig. 10 and Fig. 11, respectively. Local minima corresponding to destructive interference are identified using a dip detection algorithm, which provides improved robustness against noise compared to peak-based methods. The free spectral range (FSR) is then computed as the wavelength spacing between adjacent dips.

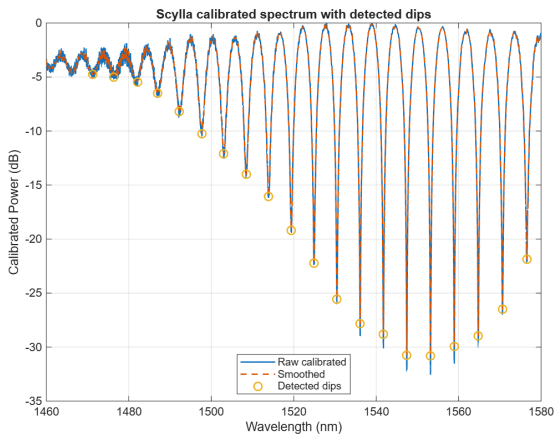

Figure 10. Calibrated spectra with detected dips for MZI4 measured on Scylla.

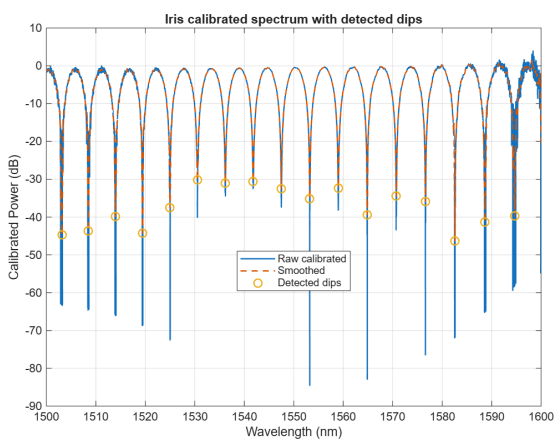

Figure 11. Calibrated spectra with detected dips for MZI4 measured on Iris.

The extracted FSR as a function of wavelength is shown in Fig. 12. A gradual decrease of FSR with increasing wavelength is observed, consistent with normal waveguide dispersion. The smooth and continuous trend indicates single-mode operation with negligible higher-order mode contributions. Within 1500nm to 1600nm wavelength window, good agreement between Scylla and Iris further confirms the consistency and robustness of the measurement and extraction procedure.

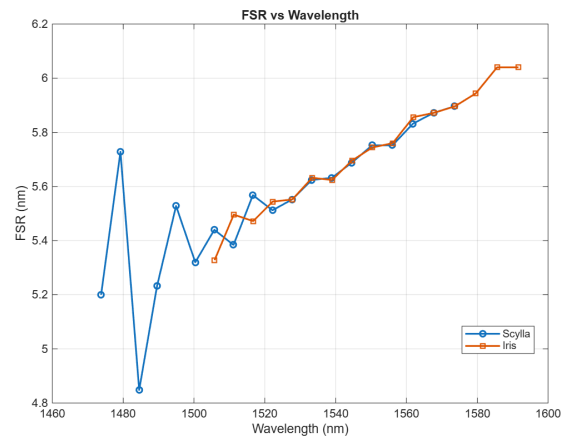

Figure 12. Extracted FSR versus wavelength for MZI4 measured on Scylla and Iris.

D. Extracted FSR and n_g vs. path length difference

Three MZI devices (MZI4, MZI5, and MZI6) with waveguide width of 500 nm and path length differences of $\Delta L = 100$, 200, and 400 μm are analyzed. The calibrated transmission spectra from Scylla and Iris are shown in Fig. 13 and Fig. 14, respectively.

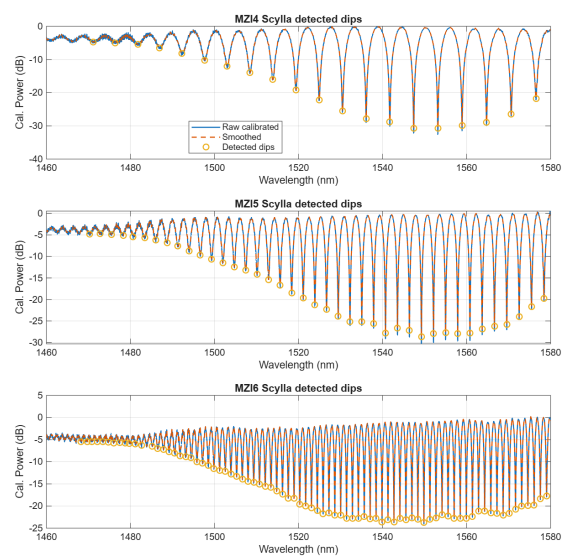

Figure 13. Measured spectra for multiple MZIs on Scylla.

The extracted FSR as a function of wavelength is shown in Fig. 15. As expected, the FSR decreases with increasing ΔL, confirming the inverse relationship

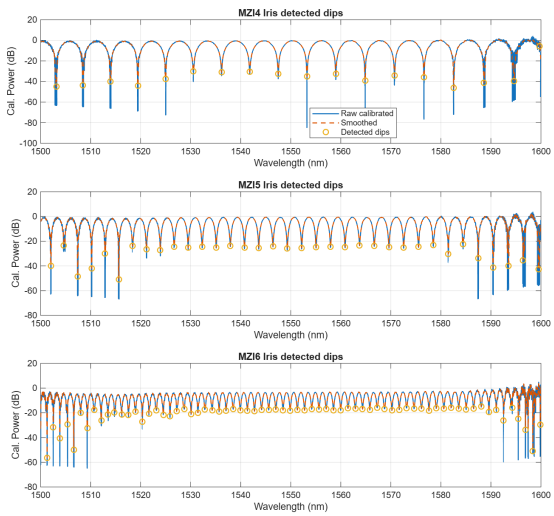

Figure 14. Measured spectra for multiple MZIs on Iris.

$FSR \propto 1/\Delta L$. This trend is consistently observed across both measurement platforms.

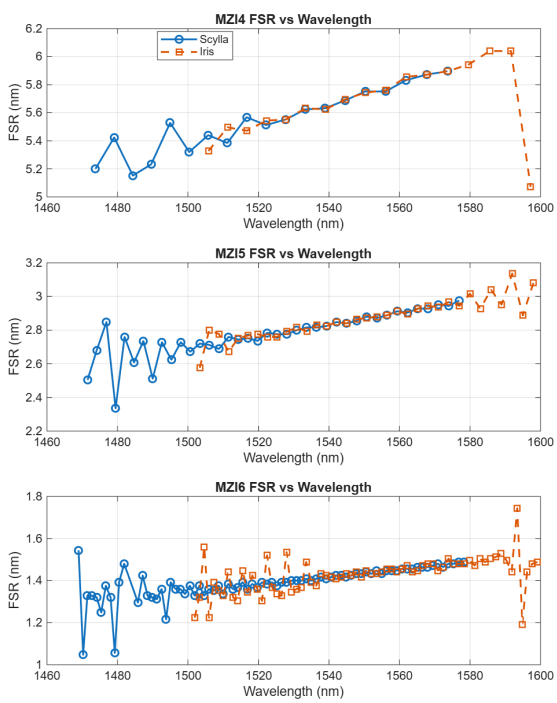

Figure 15. FSR versus wavelength for different ΔL values, demonstrating inverse scaling with path-length difference.

Using the extracted FSR values, the group index n_g is calculated based on Eq. (16), and plotted in Fig. 16. The extracted n_g values are in good agreement across all devices, indicating that the method is robust and independent of the chosen path length difference. Minor deviations between devices and from simulation are

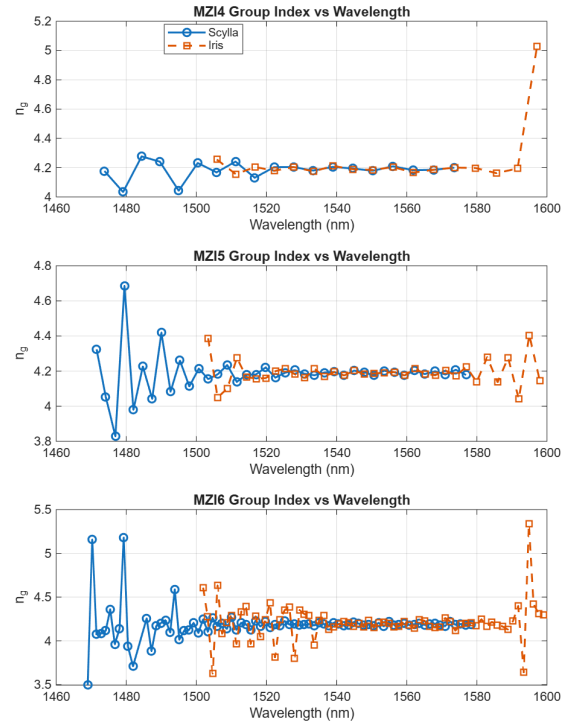

Figure 16. Extracted group index n_g versus wavelength for different MZI lengths.

attributed to fabrication tolerances, waveguide dimension variations, and measurement noise.

The linear fitting approach is shown in Fig. 17 at 1550nm. The strong linearity confirms the theoretical model. This method reduces sensitivity to local noise and provides a more reliable estimate of n_g by leveraging multiple devices. Same procedure is done at all wavelengths to extract a robust result of n_g from the linear fit, shown in Fig. 18.

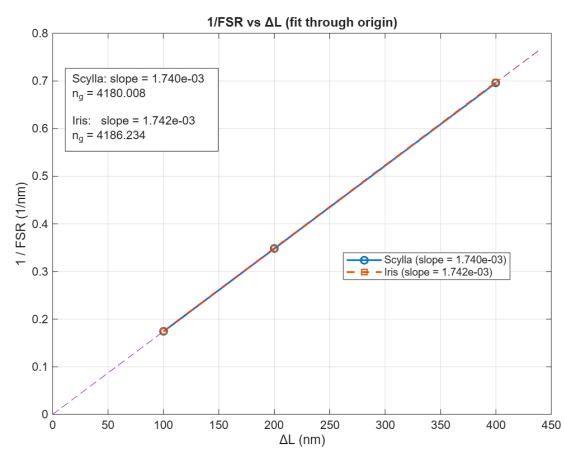

Figure 17. Linear fit of $1/\mathrm{FSR}$ versus ΔL at 1550 nm.

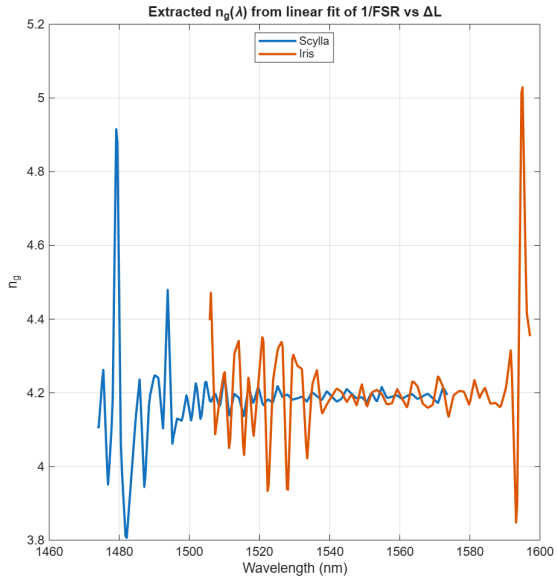

Figure 18. Linear fitting across wavelengths for robust extraction of n_g.

E. Extracted FSR and n_g vs. Waveguide Width

Three Mach–Zehnder interferometers (MZI1, MZI4, and MZI7) with waveguide widths of 400 nm, 500 nm, and 600 nm, respectively, and a fixed path length difference of $\Delta L = 100$ μm are analyzed. Using the method described in the previous subsection, the free spectral range (FSR) is extracted from the measured spectra, and the corresponding group index n_g is calculated. The extracted group index at 1550 nm is plotted as a function of waveguide width in Fig. 19 and compared with the simulated values obtained from the compact model in Table II. As the waveguide width increases, the measured n_g decreases monotonically, consistent with the expected reduction in modal confinement and dispersion.

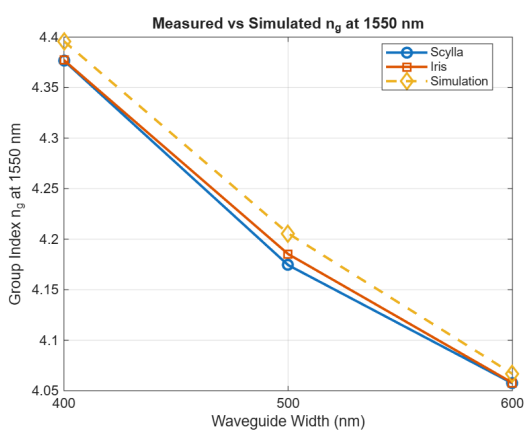

Figure 19. Measured and simulated group index n_g at 1550 nm as a function of waveguide width for MZI1, MZI4, and MZI7.

VI. Conclusion

In this work, unbalanced silicon Mach–Zehnder interferometers were designed, fabricated, and experimentally characterized to extract the waveguide group index from measured free spectral range (FSR). A systematic design-of-experiments spanning waveguide widths and path length differences enabled validation of the theoretical relationship between FSR, ΔL, and n_g. Measured spectra from two independent automated test platforms showed strong agreement, confirming the robustness of the measurement and normalization procedures. The extracted FSR exhibited the expected inverse scaling with ΔL and wavelength dependence due to dispersion, while the resulting group index values were consistent across devices and closely aligned with simulation.

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